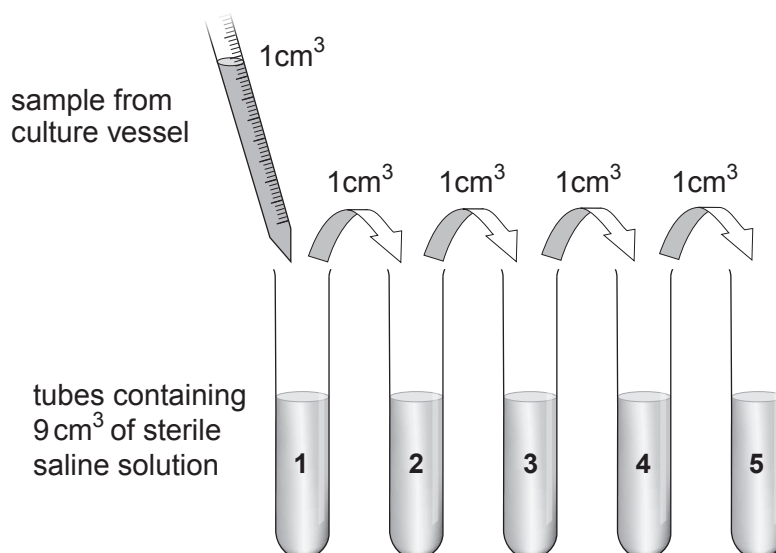


5. A large-scale culture of *Escherichia coli* was set up in an electronically maintained fermenter. Every ten minutes a sample was removed and serially diluted using the method shown in **Image 5.1**.

A 0.5 cm^3 sample from each tube was then plated onto separate sterile agar plates using aseptic techniques. The plates were incubated at 25°C for 48 hours and the number of colonies counted.

Image 5.1



- (a) (i) Describe **two** precautions that should have been taken to prevent contamination of the 0.5 cm^3 samples **as they were transferred to the agar plates**. [2]

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Examiner
only

The results for the 160-minute sample and the 240-minute sample are shown in **Table 5.2**.

Table 5.2

Sample time / min	Number of colonies counted				
	Tube 1	Tube 2	Tube 3	Tube 4	Tube 5
160	Too many to count	Too many to count	>100	32	4
240	Too many to count	Too many to count	Too many to count	>100	32

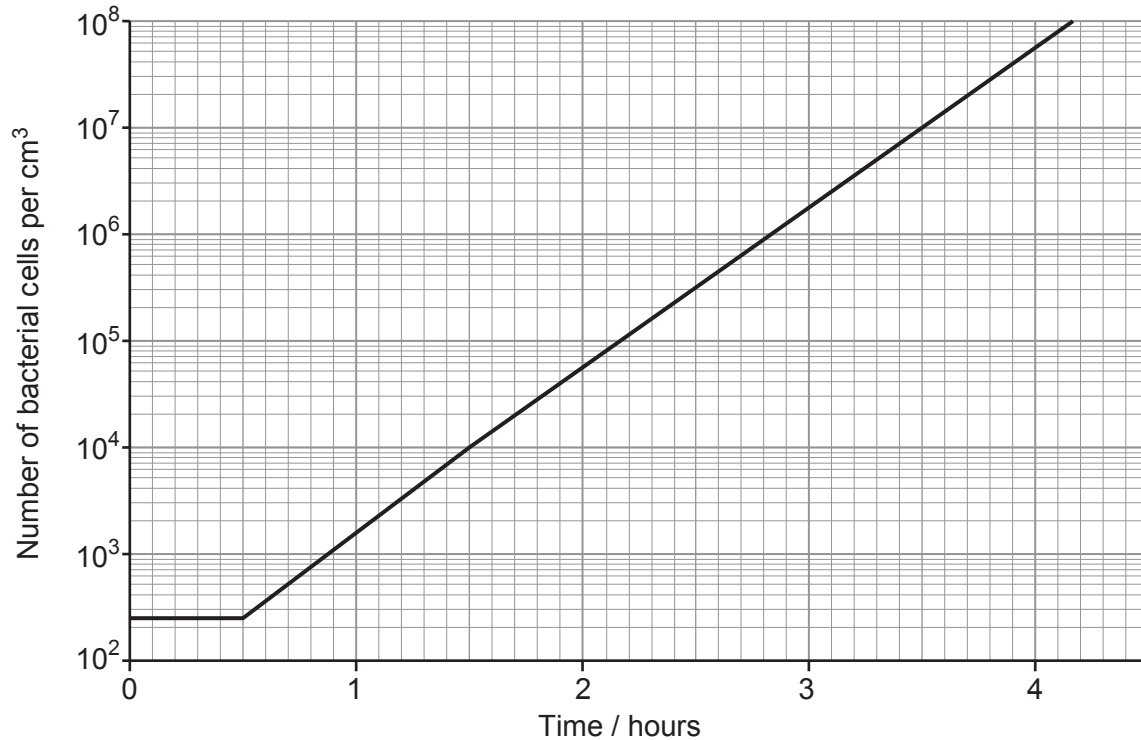
- (ii) Use **all** the information provided to calculate the number of bacterial cells **per cm³** in the culture at **160 minutes**. **Express your answer in standard form.** [2]

Number of bacterial cells = per cm³



During the experiment, the number of bacterial cells was also monitored by continuously measuring the optical density (cloudiness) of the culture using a colorimeter. **Graph 5.3** shows the population growth curve in the fermenter over a 4-hour period using the optical density method.

Graph 5.3



- (b) (i) Use **Graph 5.3** and the formula given below to calculate the number of generations produced per hour between **1.5 hours** and **3.5 hours**.

Express your answer to the nearest whole number.

[3]

$$\text{Number of generations per hour} = \frac{\log_{10} [X_t] - \log_{10} [X_0]}{0.301 \times t}$$

Where:

X_t = number of bacterial cells per cm³ at the end of the growth period

X_0 = number of bacterial cells per cm³ at the start of the growth period

t = length of growth period in hours

Number of generations per hour =



- (ii) Explain the shape of the growth curve over the first 30 minutes. [1]

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- (iii) For the first 40 minutes, the number of cells per cm^3 calculated from the agar plates as outlined in **Image 5.1** was not significantly different from the number of cells per cm^3 calculated using optical density. However, as time passed the number calculated from the agar plates was significantly lower than the number calculated using optical density.

Explain why there was no difference between the values from the two methods over the first 40 minutes but an increasing difference over the remaining time. [3]

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